Aishwarya Ramanath Shanbhag

+1 (408) 821-3391 - aishwaryaramanath.shanbhag@sjsu.edu - LinkedIn - GitHub - San Jose, CA

EDUCATION

San Jose State University — **San Jose, CA**
Master of Science in Computer Science (In Progress – Semester 1) — *Aug 2026 – May 2028*
Relevant Coursework: Topics in Machine Learning, Database Systems Principles

PES University — **Bengaluru, India**
Bachelor of Technology in Computer Science & Engineering; GPA: 3.6/4.0 — *Aug 2018 – May 2022*
Relevant Coursework: Machine Intelligence, Natural Language Processing, Big Data, Data Analytics

TECHNICAL SKILLS

Programming Languages: Python (Pandas, NumPy, Scikit-learn, PyTorch), SQL, R
ML & Statistical Modeling: Supervised & Unsupervised Learning, Predictive Modeling, Feature Engineering, Ensemble Techniques (XGBoost), Anomaly Detection, NLP, A/B Testing, Model Validation, Statistics & Probability, Hypothesis Testing, Regression Analysis, Optimization, Data Mining
Generative AI: LLMs, RAG, Prompt Engineering
Risk & Compliance Analytics: Fraud Detection, Financial Crime Risk Rating, AML/KYC, Regulatory Compliance, MRM, Stakeholder Management
Data Engineering, Cloud & BI: ETL Pipelines, Spark, Snowflake, AWS, SageMaker, QuickSight, Tableau, Git, GitHub

WORK EXPERIENCE

NatWest Group — **Bengaluru, India**
Data Scientist – Machine Learning, Fraud Detection & Prevention Squad — *Jun 2025 – Jul 2026*

- Optimized and modified the **TMX ThreatMetrix** onboarding classification model, deploying algorithmic updates that prevented **$94,000 in monthly fraud losses**, reduced **false positives by 58%**, and increased **fraud capture rates by 143%**.
- Designed **model monitoring dashboards** in **Amazon QuickSight**, embedding financial KPIs and **feature distribution shifts** for continuous algorithmic stability.
- Executed **model evaluations**, **silent period testing**, and **bias analyses** pre-deployment to assess **model drift** and fairness with cross-functional fraud strategy teams.

Societe Generale — **Bengaluru, India**
Associate Data Scientist – Compliance Data Science & Financial Crime Client Risk Rating — *Aug 2022 – May 2025*

- Led validation of **FCCR V3 Ongoing Monitoring models** as part of **Model Risk Management (MRM)** review, ensuring regulatory compliance and algorithmic integrity for **15 million+ retail banking clients**.
- Engineered **Nested Dichotomized Logit** and **Boosted C5.0** models, executing **3 monitoring cycles** across **8+ KPIs** including **PSI** to catch **data drift**, contributing to a **EUR 2 Million reduction** in regulatory penalties.
- Built a **BERTopic NLP pipeline** to automate **KYC** comment analysis, eliminating **2 FTEs** of manual review.

Hewlett Packard Enterprise (HPE) — **Chennai, India**
OCA Tool Developer Intern — *Feb 2022 – Aug 2022*

- Shipped **OCA** platform features (search, cloning, reusable templates), cutting **configuration time by 25%**.
- Built **breakout, filtering, and attribute-selection modules** and validated **Aruba-specific** product configurations.

PROJECTS

Generative Data Augmentation for Malware Family Classification (PyTorch, VAE, GAN) *Nov 2026 – Present*

- Building an **MLP classifier** on the **BODMAS** dataset (**134K samples**, 2,381 features, **581 families**) to test whether **VAE** and conditional **GAN** synthetic data improves rare-family detection.
- Designed a **leakage-controlled evaluation** with family-held-out and chronological splits, benchmarking generative augmentation against class-weighting and **SMOTE** across 5 random seeds.
- Evaluating **macro-F1**, rare-family recall, unseen-family recall at 1% FPR and robustness to feature noise.

MicroNN: Filtered Vector Search in an Embedded Vector Database (Python, SQLite) *Sep 2026 – Nov 2026*

- Re-implemented Apple's **MicroNN** vector database in a team of 3 and benchmarked it on **GloVe (1.18M vectors)**, cutting query memory **90–181x** (8.7 MB vs 1,575 MB) compared with an in-memory index.
- Designed an **adaptive filtered-search plan** that estimates selectivity mid-query, raising the share of queries hitting **0.9 recall** from 60% to **97%** with zero tuning and **20% lower latency** than pre-filtering.
- Validated 8 core paper claims on GloVe and found that the paper's query optimizer under-performs on poorly clustered data.